

Supplementary Material

Tipping the Climate Equilibrium: Tensor-Based Game Theory for Identifying Critical Coalitions in Climate Policy Negotiations

Appendix A: Final Coalitions Learned by Reinforcement Learning

The table below summarizes the final coalitions discovered by the reinforcement learning agent trained under different reward models. These were obtained after 1000 training epochs using a REINFORCE policy gradient method.

Table 1: RL-learned coalitions by reward strategy		
Reward Function	Final Coalition	Size
Raw Reward	[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]	10
Penalised Reward	[1, 2, 3, 4, 6]	5
Threshold-Based Reward	[1, 3, 5, 8]	4

These outcomes reflect the effect of reward shaping on learned policy structure. Threshold-based rewards reliably result in small tipping coalitions. Penalised rewards can balance efficiency and size when tuned appropriately, while raw rewards predictably encourage maximal inclusion.

Appendix B: Cluster Composition and Internal Similarity

To support the analysis of country groupings derived from co-ratification patterns, we include visualisations for each of the eight clusters identified in Section 6 of the main manuscript. These plots illustrate the internal diversity and cohesion of each cluster across three perspectives:

- **PCA projections** show the relative similarity of countries within the cluster in two-dimensional space;
- **Dendrograms** expose the hierarchical relationships among members and suggest potential sub-clusters;
- **Cosine similarity heatmaps** provide a direct comparison of ratification overlap within the group.

Each visualisation is labelled using the compact descriptive cluster names introduced in Section 6. Cluster 5, which contains a notably large number of countries, demonstrates internal variation suggestive of three subgroups. These could be resolved in future work using finer-grained clustering methods or application-specific criteria.

The full country-to-cluster assignment table, including descriptive cluster labelling, is available at the project repository:

[details withheld for double-anonymous review]

Full cluster figures (PCA, dendrogram, heatmap) are provided in the supplementary materials folder: `figures/clusters/`.

Appendix C: Greedy Algorithm Validation

To verify that the greedy forward-selection pipeline computes scores consistently with the exhaustive brute-force search, we conducted a direct score-consistency check. A stratified sample of 1,500 coalitions (500 per size class, sizes 2–4) was drawn from the full brute-force reference of 38,646,125 coalitions. For each sampled coalition, the *TippingScore* was recomputed independently using the pipeline matrices (`cosine_similarity.npy`, `influence_matrix.npy`, cluster assignments), then compared to the stored reference value.

All 1,500 recomputed scores matched the reference exactly to floating-point precision: mean absolute error = 0, maximum relative error = 2.35×10^{-16} (machine epsilon). This confirms that both the exhaustive search and the greedy algorithm use the same scoring formula and that no numerical drift was introduced. Figure 1 shows the scatter of recomputed versus reference scores; all points lie exactly on the diagonal.

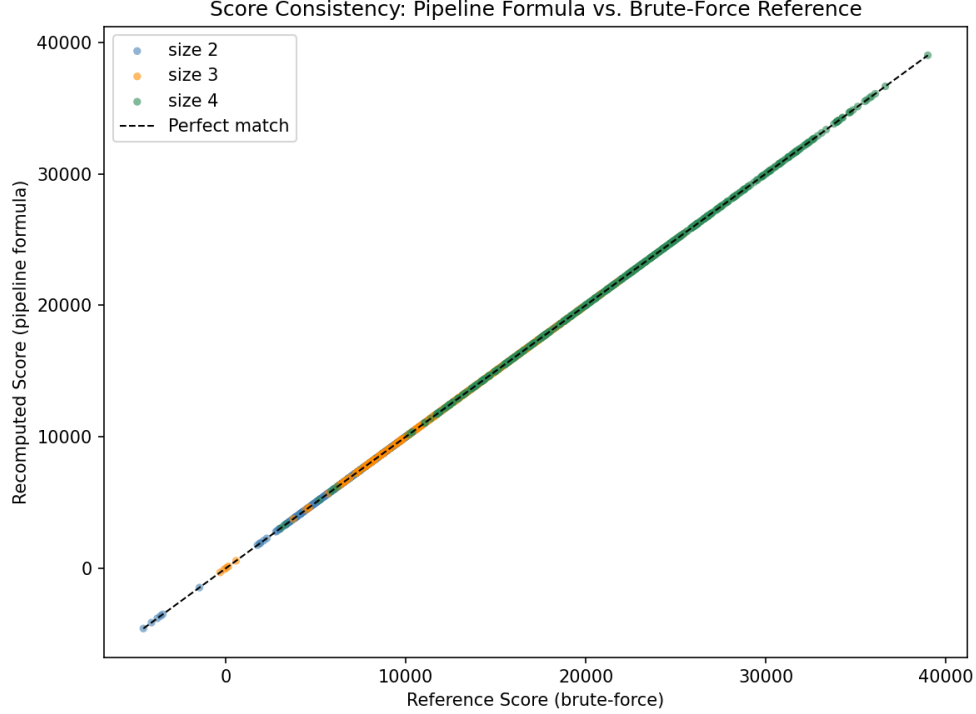


Figure 1: Score consistency check: recomputed *TippingScore* (pipeline formula) versus stored brute-force reference score for a stratified sample of 1,500 coalitions (sizes 2–4). All points fall exactly on the diagonal (max relative error 2.35×10^{-16}), confirming numerical equivalence between the exhaustive and greedy scoring implementations.

These results confirm that the scoring formula is implemented consistently across both the exhaustive brute-force enumeration and the greedy forward-selection pipeline. The greedy method’s ability to identify high-scoring coalitions without full enumeration makes it practical for candidate discovery in large combinatorial spaces (Leskovec et al., 2014).

References

Leskovec, J., Rajaraman, A., and Ullman, J. D. (2014). Mining of massive datasets. *Cambridge University Press*. Chapter 13: Link Analysis and Greedy Algorithms.